

Adverse selection and screening III

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MSPC Chapter 4.3-4.5

Introduction

Generic adverse selection/screening problem:

- max expected output (expectation taken w.r.t. nature's probability distribution)
- s.t. PCs
- and s.t. ICs
- More general IC when any number of types:

$$e \in \arg \max_{\hat{e}} \left\{ \sum_{i=1}^n p_i(\hat{e}) u(w(x_i)) - v(\hat{e}) \right\}$$

Introduction

- We now return to adverse selection in labour markets, where a worker may be better informed about his innate ability (productivity) than his employer.
- To have a basis for our signalling model we consider a model similar to that in the hand-in problem in PS10.
 - continuum of types (as in chapter 4.4. But there are discrepancies.)
 - multiplicity of equilibria may arise
 - no screening mechanism (instead we will introduce a signalling mechanism later)

Model

- Many identical firms that try to hire workers (competitive market).
- Each firm's production technology exhibits CRS and uses only labour.
- Price of output is 1. Firms act as output price takers.
- There is a continuum of workers with mass N .
- Workers differ in their productivity $\theta \in [\underline{\theta}, \bar{\theta}]$. If a worker of type θ works for a firm, he produces θ units of output.
- The distribution of θ represented by c.d.f. $F(\theta)$, with density $f(\theta) > 0$ for all θ . Each worker knows his own productivity.
- A type θ worker's outside option is $r(\theta)$.
- Workers are earnings maximizers.

- A firm's Bernoulli utility function is

$$\Pi(\theta, w) - w.$$

So firms maximize expected profit.

- Not having a screening mechanism eliminates the equilibrium existence problem.

Efficient allocation

- To find the Pareto efficient allocation of workers write down the aggregate surplus:

$$N \int_{\underline{\theta}}^{\bar{\theta}} [I(\theta)\theta + (1 - I(\theta))r(\theta)]f(\theta)d\theta$$

where $I(\theta) = 1$ if type θ worker is employed and 0 otherwise.

- This function is maximized by setting $I(\theta) = 1$ iff $\theta \geq r(\theta)$ and $I(\theta) = 0$ otherwise. Hence, in any Pareto efficient allocation the set of workers employed must be

$$\{\theta : r(\theta) \leq \theta\}$$

First best competitive equilibrium

Suppose worker's productivity level is publicly observable.

- Because the labour of each different type of worker is a distinct good, there is a distinct equilibrium wage $w^*(\theta)$ for each type θ .
- All firms offer wages $w^*(\theta) = \theta$ to workers of type θ , and workers accept one of these wage offers at random.
- Competition drives wages to marginal product.
- So type θ works if and only if $r(\theta) \leq w^*(\theta) = \theta$.
- This market allocation is efficient.
- Separate markets for distinct goods.

Asymmetric information

Firms don't know θ , workers do.

- If productivity is unobservable the wage cannot depend on productivity. Why not just ask workers?
- There must be a single wage w for all workers.

Labour supply:

- A worker is willing to work for a firm iff

$$r(\theta) \leq w$$

- Thus the set of worker types that want to work is given by

$$\Theta(w) = \{\theta : r(\theta) \leq w\}.$$

Asymmetric information

Labour demand:

- Let μ denote firms' belief (or expectation) about average productivity.
- Labour demand is given by (why?)

$$z(w) = \begin{cases} 0 & \text{if } \mu < w \\ [0, \infty] & \text{if } \mu = w \\ \infty & \text{if } \mu > w \end{cases}$$

- We require firms to have correct beliefs:
 $\mu(w) = E[\theta | \theta \in \Theta(w)]$. I.e. $\mu(w)$ is firms' expectation about the average productivity of those showing up for work when the wage is w .
- Note: Demand equals supply iff

$$\mu(w) = w$$

Equilibrium

Definition: In the competitive labour market model with unobservable worker productivity levels, a competitive equilibrium is a wage w^* and a set Θ^* of worker types who accept employment such that

$$\Theta^* = \{\theta | r(\theta) \leq w^*\}$$

$$w^* = E[\theta | \theta \in \Theta^*]$$

Comments:

- The equilibrium requires that firms form rational expectations about which workers are going to accept employment.
- If Θ^* is empty, we assume that firms use the unconditional expectation, so $E[\theta | \theta \in \emptyset] = E(\theta)$.
- If we instead assume different “out of equilibrium beliefs” it can give rise to other equilibria

A competitive equilibrium with asymmetric information need not be Pareto efficient.

Example 1:

Suppose that $r(\theta) = r$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$, with $F(r) \in (0, 1)$.

Consider a competitive equilibrium:

- 1 If $w \geq r$, all workers accept employment. $\Theta^*(w) = [\underline{\theta}, \bar{\theta}]$
- 2 If $w < r$, no worker accepts employment. $\Theta^*(w) = \emptyset$
 - In either case, $E[\theta | \theta \in \Theta(w)] = E(\theta)$.
 - Thus, the equilibrium wage is $w^* = E(\theta)$.
 - Both situations 1 and 2 are inefficient! Why?

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If $E(\theta) \geq r$, that is, if there are enough workers with a relatively high productivity, we are in situation 1. If $E(\theta) < r$, i.e., if there are enough workers with a relatively low productivity, we are in situation 2.

Equilibrium is inefficient because:

Pareto efficiency requires that workers accept employment if and only if $\theta \geq r$.

We either get too much work and over production in firms:

- if $w^* \geq r$ and all workers accept employment
- also the workers for which $\theta < r$, who actually "produce better at home".

Or we get too little work (actually none) and no production in firms:

- if $w^* < r$ and no worker accepts employment
- workers who should contribute by working at firms do not work.

Asymmetric information and market unravelling

- If $r(\theta)$ varies with θ the market may break down completely, i.e., no worker will be hired by any firm even though it would be efficient that all workers are employed.
- To see this suppose $r(\theta) \leq \theta$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. I.e., every worker is (weakly) more efficient working for a firm than working at home.
- Assume also that $r'(\theta) > 0$. I.e., workers who are more productive working in a firm are also more productive working at home.

As argued above, the equilibrium wage w^* must satisfy

$$w^* = E[\theta | r(\theta) \leq w^*]$$

We can illustrate in a $(w, \text{productivity})$ -diagram. Notes for drawing the diagram:

- 45-degree line: mapping between types and break-even wage.
- $E[\theta | r(\theta) \leq w]$ must be increasing in w . Why?
- $E[\theta | r(\theta) \leq r(\underline{\theta})] = \underline{\theta}$. I.e., if the wage is equal to the outside option of the least productive worker, then only this least productive worker is willing to accept employment.
- If $r(\underline{\theta}) < \underline{\theta}$, the function $E[\theta | r(\theta) \leq w]$ starts above the 45-degree line.
- If $r(\bar{\theta}) > E[\theta]$, the function $E[\theta | r(\theta) \leq w]$ must intersect the 45-degree line.

- $E[\theta|r(\theta) \leq r(\bar{\theta})] = E[\theta]$. I.e. if wage is equal to the outside option of the most productive worker, then every worker is willing to accept employment. Thus expected productivity equals the unconditional expected productivity. Same if wage is greater than $r(\bar{\theta})$.
- In a competitive equilibrium, we have $w^* = E[\theta|r(\theta) \leq w^*]$. Thus, any intersection of $E[\theta|r(\theta) \leq w]$ and the 45-degree line (i.e. any fixed point of the function) is an equilibrium.
- At w^* the set of workers with $r(\theta) \leq w^*$ has expected productivity $E[\theta|r(\theta) \leq w^*] = w^*$.

- The market equilibrium (illustrated in diagram) is in efficient when $E(\theta) < r(\bar{\theta})$.
- To get the best workers to accept employment, the wage need to be at least $r(\bar{\theta})$. But then firms cannot break even if $E(\theta) < r(\bar{\theta})$.
- The presence of lower types forces the wage down, which in turn drives the best workers out of the market.
- But once they are driven out of the market the average productivity falls and so on.
- So it is the high θ workers who are not working.
- This is the phenomenon of **adverse selection**.

How far can this process go?

- It may easily happen that the market unravels completely and that no trade takes place.
- In illustration $r(\underline{\theta}) = \underline{\theta}$ and $r(\theta) < \theta$ for all other types.
- So $w^* = \underline{\theta}$ and only type $\underline{\theta}$ accepts employment. Essentially, no worker is hired.
- Problem: The presense of low productivity workers keeps expected productivity low.

Specific example of market unravelling

- Let $r(\theta) = \alpha\theta$, $\alpha \in (\frac{1}{2}, 1)$, $\theta \sim U[0, 2]$.
- Thus, $r(\underline{\theta}) = 0$.
- Solve: $E[\theta | r(\theta) \leq w] = E[\theta | \alpha\theta \leq w] = \frac{w}{2\alpha}$
- So $E[\theta | r(\theta) \leq w^*] = w^* \Leftrightarrow w^* = \frac{w^*}{2\alpha} \Leftrightarrow w^* = 0$.
- $\Theta^* = \{\underline{\theta}\}$

It is also possible to get multiple equilibria:

- The slope of $E[\theta|r(\theta) \leq w]$ depends on the distribution.
Depends on the density of workers who are indifferent about accepting employment or not.
- If in some range the density is high, then the slope of $E[\theta|r(\theta) \leq w]$ is steep. If in some range the density is low, then it is flat.
- These equilibria can be Pareto ranked:
 - Firm's profits must be 0 in any competitive equilibrium. Why?
 - All workers prefer higher wages to lower wages.
 - Equilibria with higher wages Pareto dominate those with lower ones

Efficiency

In the case where $r(\theta) < \theta$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$:

- Only if everybody is employed in the highest wage competitive equilibrium the outcome is fully efficient.
- Otherwise the outcome is inefficient.

Reminder:

An allocation is “constrained Pareto optimal” or “second best optimal” if it cannot be Pareto improved by a social planner who is unable to observe the agents private information.

Comments:

- An allocation that is Pareto optimal is clearly also constrained Pareto optimal
- In the labour market example all equilibria except for the highest wage equilibrium are not constrained Pareto optimal. A social planner could achieve a Pareto improvement by implementing the allocation of the highest wage equilibrium: Set $w = w^*$ and let workers choose whether to accept employment or not.

- The highest wage equilibrium is constrained Pareto optimal. Without knowing the workers types the social planner cannot make non-workers (and workers) better off without hurting firms. (furthermore in this model there is no way to screen or signal)
- The government could raise total social surplus by raising the wage to $w = E(\theta)$ and forcing all workers to accept employment. However, this is not a Pareto improvement because the high productivity workers are made worse off.
- Such a policy may be justified from an ex ante perspective. If workers were asked before learning their productivities (behind the “veil of ignorance”) whether they would agree to such a policy, all workers should unanimously agree to it.